

Solar Power

Photovoltaic panels convert sunlight directly into electricity using semiconductor cells, usually made of silicon. When photons strike a cell they free electrons, creating a current. Panel efficiency for commercial modules is typically between eighteen and twenty-three percent. Output depends on irradiance, panel angle, temperature and shading, so installers orient arrays towards the equator and tilt them close to the local latitude.

Wind Power and Storage

Wind turbines capture the kinetic energy of moving air with rotor blades that drive a generator. Power rises with the cube of wind speed, which is why turbines are placed offshore or on exposed ridges where winds are strong and steady. Because both wind and solar generation are intermittent, grids rely on storage such as lithium-ion batteries and pumped hydro to shift energy from periods of surplus to periods of high demand.